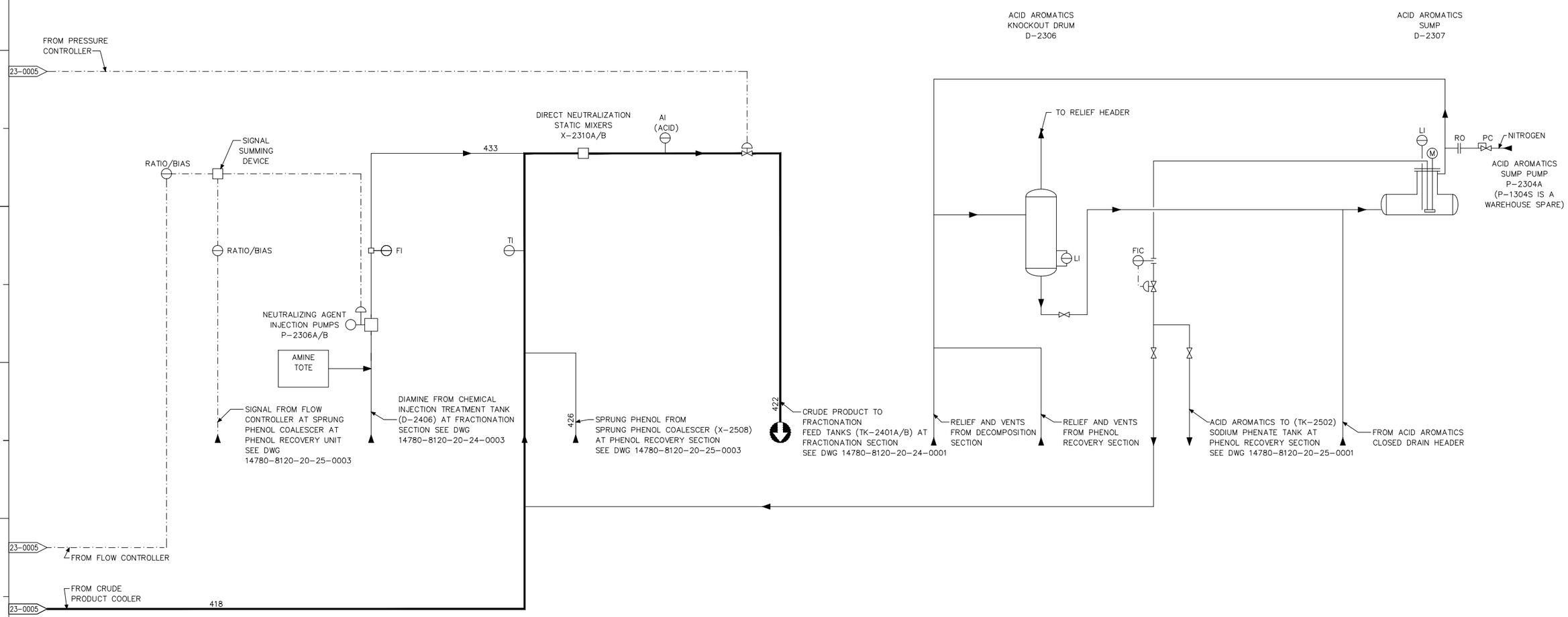
14780-8120-20-23-0006



STREAM DATA SUMMARY				
STREAM NUMBER	418	422	426	433
TEMPERATURE, deg C	43	43	41	31
PRESSURE, kg/cm2[g](mm Hg)	----	----	----	----
TOTAL MASS FLOW, kg/h	61258	63255	1994	2.8

MATERIAL BALANCE, kmol/h

STREAM	151	152	229	230	231	236	299	329	340	345	346	350	355	360	377	378	401	422	426	433	475
WATER	----	----	16.20	16.20	3.40	0.10	----	----	12.58	0.03	78.68	0.07	0.07	94.40	20.81	----	TRACE	38.97	5.53	----	----
NITROGEN	0.01	0.01	7.57	7.57	1.50	0.04	0.17	0.07	----	TRACE	----	TRACE	2.98	TRACE	----	0.27	----	0.27	----	----	1.29
OXYGEN	----	----	0.51	0.51	0.10	TRACE	0.04	0.02	----	TRACE	----	TRACE	0.16	TRACE	----	----	----	----	----	----	----
FORMIC	----	----	0.89	0.89	0.19	TRACE	----	----	----	0.01	----	0.04	TRACE	0.14	----	----	----	TRACE	----	----	----
METHANOL	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	0.06	----	----	----
ACETONE	----	----	0.16	0.16	0.02	TRACE	----	----	----	TRACE	----	0.01	TRACE	0.01	----	----	----	335.02	0.04	----	----
ACETOL	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	0.36	----	----	----
PHENOL	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	344.85	11.05	----	----
MO	----	----	TRACE	TRACE	TRACE	TRACE	----	----	----	TRACE	----	TRACE	TRACE	TRACE	----	----	----	0.03	----	----	----
ACP	----	----	7.20	7.20	1.53	0.05	----	----	----	TRACE	----	0.44	TRACE	TRACE	----	----	----	1.20	----	----	----
AMS	----	----	0.69	0.69	0.14	TRACE	----	----	----	TRACE	----	0.12	TRACE	TRACE	----	----	----	10.96	0.01	----	----
AMSO	----	----	3.44	3.44	0.71	0.02	----	----	----	TRACE	----	0.32	TRACE	TRACE	----	----	----	0.36	----	----	----
CUMENE	27.82	27.82	6831.99	6831.99	1294.52	38.84	----	----	----	22.26	----	1168.84	0.01	0.69	----	----	----	70.96	7.07	----	----
DMPC	----	----	62.94	62.94	13.74	0.41	----	----	----	TRACE	----	1.38	TRACE	TRACE	----	----	----	0.36	----	----	----
U.I.RES.	----	----	2.44	2.44	0.48	0.01	----	----	----	TRACE	----	0.05	TRACE	TRACE	----	----	----	0.70	----	----	----
CHP	----	----	1600.81	1600.81	346.38	10.39	----	----	----	TRACE	----	0.01	TRACE	TRACE	----	----	----	TRACE	----	----	----
PH.RESIN	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	0.09	----	----	----
PCP	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	1.44	----	----	----
BPA	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	TRACE	----	----	----
AMS.DIM1	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	0.35	----	----	----
DCP	----	----	2.01	2.01	0.44	0.01	----	----	----	TRACE	----	TRACE	TRACE	TRACE	----	----	----	0.04	----	----	----
CPCHP	----	----	1.18	1.18	0.26	TRACE	----	----	----	TRACE	----	TRACE	----	----	----	----	----	TRACE	----	----	----
RESIDUE	TRACE	TRACE	0.23	0.23	0.04	TRACE	----	----	----	----	----	----	----	----	----	----	----	0.04	----	----	----
DICUPH	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	TRACE	----	----	----
HMDA	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	0.02	----	0.02	----
H2SO4	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	0.02	0.02	TRACE	----	----
MOLE FLOW, kmol/h	27.84	27.84	8538.25	8538.25	1663.45	49.90	0.21	0.09	12.58	22.31	78.68	1171.28	3.23	95.24	20.81	0.27	0.03	806.14	23.71	0.02	1.29
MASS FLOW, kg/h	3346	3346	1076643	1076643	210868	6326	6.1	2.7	227	2677	1417	140798	92	1791	375	7.5	2.5	63255	1994	2.8	36
MOLE WEIGHT	120.17	120.17	126.10	126.10	126.77	126.77	28.85	28.85	18.02	120.03	18.02	120.21	28.38	18.80	18.02	28.01	90.07	78.47	84.08	116.21	28.01
LIQ STD VOL FLOW, m3/h	3.87	3.87	1195.65	1195.65	233.15	6.99	0	0	0.23	3.10	0	162.85	0	1.76	0.38	0	1.362E-3	67.00	2.04	3.355E-3	0
LIQ STD DENSITY, kg/m3	865.2	865.2	901.4	901.4	905.4	905.4	0	0	999.1	865.4	0	865.5	0	995.7	999.1	0	1800.6	945.1	976.3	848.8	0
VAP STD VOL FLOW, norm m3/h	0	0	0	0	0	0	4.7	2.1	0	0	1764	0	72	7.0	0	6.0	0	0	0	0	29
VAP STD DENSITY, kg/m3	0	0	0	0	0	0	1.2	1.2	0	0	0.76	0	1.2	5.0	0	1.2	0	0	0	0	1.2

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REV.	DESCRIPTION	BY	CHK.	APPR.	PROJ. APPR.	DATE
Z1	AS-BUILT	HSO	SWP	JSC	SWL	4/25/16
S1	ISSUED FOR CONSTRUCTION NO HOLDS	HSO	SWP	JSC	SWL	4/25/14
A1	APPROVED FOR DESIGN	HSO	SWP	JSC	SWL	9/12/13
01	COMMENT ISSUE	HSO	SWP	JSC	SWL	7/11/13

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BASED ON 963766-110-06-A1

PROCESS FLOW DIAGRAM
NEUTRALIZING AGENT INJECTION –
KNOCKOUT DRUM-ACID AROMATICS SUMP NEUTRALIZATION SECTION
CONCENTRATION, DECOMPOSITION AND NEUTRALIZATION SECTION
UOP PHENOL PROCESS UNIT

CERTIFIED FOR CONSTRUCTION		DWG.	DSGN.	CHK.	APPR.
BY	BY				
DATE	DATE				

PROJ. NO.	DWG. NO.	REV.
PEN 120117	14780-8120-20-23-0006	Z1
PPCL		

SCALE